



Bisphenol A and alternatives in thermal paper receipts - a German market analysis from 2015 to 2017



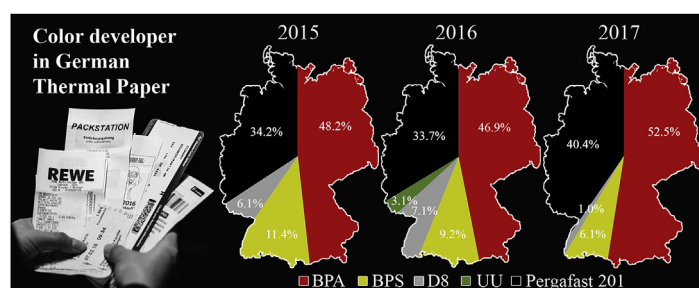
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HIGHLIGHTS

- BPA is the most commonly used color developer in 311 receipts from German market.
- Pergafast® 201 is the most frequently used BPA alternative.
- Fast degradation of Pergafast® 201 in acidic and protic solvents was observed.
- Sensitizers DPE and PMN were detected in a large number of thermal papers.
- Release experiments of color developers (skin contact, DIN EN 646) were presented.

GRAPHICAL ABSTRACT



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ABSTRACT

Bisphenol A (BPA) was commonly used as color developer for thermal paper such as cash register receipts, labels or tickets. Therefore, thermal paper was considered by the European Food Safety Authority (EFSA) as the main source of human exposure to BPA beside epoxy based food contact materials. In this study, a German market analysis on the use of BPA and alternative color developers in thermal paper receipts is provided for the years 2015, 2016 and 2017. 114 (2015), 98 (2016) and 99 (2017) samples were randomly collected and analyzed by HPLC-DAD. In summary, BPA was still the most frequently found color developer (48.2% in 2015, 46.9% in 2016 and 52.5% in 2017). The most commonly used alternative was the phenol-free substance Pergafast® 201 (34.2%, 33.7%, 40.4%). The bisphenol analogs bisphenol S (BPS; 11.4%, 9.2%, 6.1%) and D8 (6.1%, 7.1%, 1.0%) were less common. Another phenol-free substituent, a urea urethane compound (UU), was also detected (3.1% in 2016). Concentrations of color developers in thermal paper ranged from 1.4 to 32.4 mg/g (median values between 2.5 and 15.9 mg/g). Concentrations of BPA were found to be highest followed by BPS, UU, Pergafast® 201 and D8. In addition, two pharmacologically active substances, dapsone (6.0 mg/g) and tolbutamide (5.5 mg/g), were detected in a non-marketed thermal paper, that was supposed to use ascorbic acid as initial color developer. Different release experiments of the detected color developers were performed. Sensitizers 1,2-diphenoxy-ethane, 1-phenylmethoxy-naphthalene and diphenylsulfone, used frequently in the thermal paper processes, were quantified.

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